

On data-driven control of linear systems with quantized data

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Abstract—This paper studies data-driven stabilization of unknown discrete-time linear systems using only quantized state measurements that take values in finite sets. We consider a two-stage framework in which quantized data are first collected to design a stabilizing feedback gain, followed by closed-loop stabilization using quantized state feedback. For the controller design stage, we provide a verifiable condition ensuring that the obtained feedback gain is stabilizing, and establish that this condition can be guaranteed to hold under sufficiently fine quantization. For the stabilization stage, we formulate an explicit sufficient bound on the number of quantization cells that ensures exponential convergence. We characterize the trade-off between the quantization requirements in the two stages. In particular, finer quantization during controller design requires more quantization cells for data collection but improves the verifiable condition. Conversely, as the coefficient in the verifiable condition approaches zero, the sufficient bound for exponential convergence grows at least inversely with this coefficient. Numerical results illustrate the trade-off and the effectiveness of the proposed approach.

I. INTRODUCTION

Data-driven control has emerged as a powerful alternative to model-based design when accurate system models are unavailable or difficult to obtain [1], [2]. Rather than first identifying a model, direct data-driven methods construct controllers directly from measured trajectories [3], [4]. In this direction, Willems' fundamental lemma [5] shows that the behavior of a controllable linear system can be fully characterized from persistently exciting data. This key result has enabled data-driven approaches to stabilization and optimal control [6]–[8], model predictive control [9], and extensions to nonlinear systems [10] and switched systems [11].

In networked control systems, measurements are often transmitted through communication channels with limited data rates. Quantization provides a natural mechanism for representing measurements using finite alphabets. Feedback stabilization under quantization and data-rate constraints has been studied extensively; see, e.g., [12]–[15]. Recent developments span robust control [16]–[18], event-triggered control [19], and switched systems [20], [21]. These model-based results motivate incorporating data-rate constraints directly into data-driven controller design.

Robust data-driven control methods can accommodate noisy data using tools such as semidefinite programming [8], the matrix S-lemma [22], and data informativity [23]. Quantization, however, introduces an additional challenge: it can affect both the controller design and the subsequent feedback implementation.

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Only a limited number of works have investigated data-driven control under quantized measurements. In [24], unknown linear systems are stabilized using quantized control inputs together with noisy unquantized trajectory data. Other approaches consider super-stabilization from partially quantized data [25], event-triggered control with quantized state transmission [26], and the trade-off between quantization densities for data-driven design and feedback control [27]. However, these approaches rely on unquantized data for controller synthesis, employ static quantizers with infinitely many levels, or both, and therefore do not simultaneously address controller design and stabilization using only finite-valued quantized state measurements.

To address this issue, [28] introduces a two-stage framework in which quantized state measurements are first collected under a prescribed quantization error bound and used to design a feedback controller via a semidefinite program (SDP), after which a Lyapunov-based quantizer with a fixed number of quantization cells is used for feedback stabilization. Sufficient conditions were established for obtaining a stabilizing controller from the quantized data and for exponential convergence under quantized state feedback.

The present paper builds on this framework by strengthening the controller design guarantee, deriving a more explicit sufficient condition for exponential convergence, and, most importantly, characterizing the trade-off between the quantization requirements in the two stages. The main contributions are as follows.

First, for the controller design stage, we establish that, for sufficiently small quantization errors, every optimal solution to the quantized SDP satisfies a verifiable condition ensuring that the resulting feedback gain is stabilizing. Compared with [28], this extends the guarantee to the verifiable condition, at the expense of a more restrictive sufficient bound on the quantization error.

Second, for the stabilization stage, we derive from the sufficient condition for exponential convergence in [28] a more explicit bound N^* on the number of quantization cells, which also facilitates the subsequent trade-off analysis.

Third, we characterize the trade-off between the quantization requirements for controller design and subsequent stabilization. As the design-stage quantization error bound δ approaches zero, we show that the number of quantization cells per dimension N_k during data collection grows as $N_k = \Theta(\delta^{-1})$, while the coefficient η_1 in the verifiable condition approaches one with $1 - \eta_1 = O(\delta)$. Conversely, for strictly unstable plants, we show that the sufficient bound N^* for exponential convergence satisfies $N^* = \Omega(\eta_1^{-1})$ as $\eta_1 \rightarrow 0^+$, which improves to $N^* = \Theta(\eta_1^{-1})$ under additional

boundedness conditions.

The remainder of the paper is organized as follows. Section II introduces the problem formulation and quantization scheme. Sections III and IV develop the controller design and feedback stabilization stages, respectively. Section V characterizes the trade-off between their quantization requirements. Section VI numerically illustrates the proposed approach and the quantization trade-off. Section VII concludes the paper.

Notation: For a vector v , denote by $\|v\|$ and $\|v\|_\infty$ its Euclidean and infinity norms, respectively. For a matrix $M \in \mathbb{R}^{n \times m}$, denote by $\rho(M)$, $\sigma(M)$, $\|M\|$, and $\|M\|_F$ its spectral radius, smallest singular value, spectral norm (i.e., largest singular value), and Frobenius norm, respectively. For a symmetric matrix $S \in \mathbb{R}^{n \times n}$, denote by $\bar{\lambda}(S)$ and $\underline{\lambda}(S)$ its largest and smallest eigenvalues, respectively. For symmetric matrices $P, R \in \mathbb{R}^{n \times n}$, write $P \succ R$ and $P \succeq R$ when $P - R$ is positive definite and positive semidefinite, respectively.

II. PROBLEM FORMULATION

Consider a discrete-time linear time-invariant (LTI) system

$$x(k+1) = Ax(k) + Bu(k), \quad x(0) = x_0, \quad k \in \mathbb{Z}_{\geq 0}, \quad (1)$$

where $x(k) \in \mathbb{R}^n$ is the state and $u(k) \in \mathbb{R}^m$ is the control input. We assume that the dimensions n and m are known, but the matrices $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{n \times m}$ are unknown. Instead, only upper bounds on their norms are known:

$$\|A\| \leq \alpha, \quad \|B\| \leq \beta. \quad (2)$$

Our goal is to stabilize the unknown LTI system (1) using quantized state measurements $\hat{x}(k) = q_k(x(k))$, where $q_k(\cdot)$ is a time-varying quantizer that maps $\mathbb{R}^n$ to a *finite* set of points at each time k .

The proposed approach follows the two-stage framework introduced in [28]. First, in the *controller design stage*, we collect input and quantized state data and use them to compute a stabilizing feedback gain. The quantizer update rule in this stage is designed to maintain a prescribed quantization error bound. Second, in the *stabilization stage*, we employ a Lyapunov-based quantizer update rule with a fixed number of quantization cells and establish exponential convergence under quantized state feedback. The controller design stage need not be performed offline. For concreteness, we consider stabilization beginning directly after controller design without resetting the system; the designed controller can also be applied after a system reset or at a later time.

The quantization scheme is introduced in Section II-A. The design and stabilization stages are developed in Sections III and IV, respectively.

A. Quantization scheme

The initial state x_0 is unknown. We assume that there exists a known bound $E_0 > 0$ such that

$$\|x_0\|_\infty \leq E_0.$$

If unavailable, such a bound can be obtained in finite time using search techniques in quantized control; see, e.g., [12].

At each time k , we maintain a bound E_k such that

$$\|x(k)\|_\infty \leq E_k, \quad (3)$$

that is, $x(k)$ lies in the hypercube

$$\mathcal{R}(E_k) := \{v \in \mathbb{R}^n : \|v\|_\infty \leq E_k\},$$

which serves as the quantization range. Given E_k , quantizer update rules are designed to determine E_{k+1} such that $x(k+1) \in \mathcal{R}(E_{k+1})$. The update rules differ between the two stages and are derived in Sections III-B and IV-A.

The quantizer partitions $\mathcal{R}(E_k)$ into N_k^n equal hypercubic cells, with N_k cells per dimension, where N_k is a positive integer. The quantized measurement $\hat{x}(k)$ is defined as the center of the cell containing $x(k)$. Consequently, the quantization error

$$d(k) := \hat{x}(k) - x(k)$$

satisfies

$$\|d(k)\|_\infty = \|\hat{x}(k) - x(k)\|_\infty \leq \frac{E_k}{N_k}. \quad (4)$$

Since the quantizer takes values in a set of cardinality N_k^n , the measurement at time k can be represented using a finite number of bits.

III. DATA-DRIVEN CONTROLLER DESIGN

A. Data-driven control preliminaries

During the controller design stage, we excite the unknown LTI system (1) with an input sequence $\{u(k)\}_{k=0}^{T-1}$ satisfying

$$\|u(k)\| \leq \bar{u}, \quad \forall k \in \{0, 1, \dots, T-1\}, \quad (5)$$

and collect the quantized state measurement sequence $\{\hat{x}(k)\}_{k=0}^{T-1}$. Define the corresponding data matrices

$$\begin{aligned} \hat{X}_0 &:= [\hat{x}(0) \quad \hat{x}(1) \quad \cdots \quad \hat{x}(T-1)], \\ \hat{X}_1 &:= [\hat{x}(1) \quad \hat{x}(2) \quad \cdots \quad \hat{x}(T)]. \end{aligned} \quad (6)$$

The data matrices $U_0 \in \mathbb{R}^{m \times T}$, $X_0, X_1 \in \mathbb{R}^{n \times T}$, and $D_0, D_1 \in \mathbb{R}^{n \times T}$ for the input $u(k)$, unquantized state $x(k)$, and quantization error $d(k)$, respectively, are defined analogously. The matrices X_0, X_1, D_0, D_1 are introduced only for analysis and are not available for control synthesis.

A key condition for computing a stabilizing controller is

$$\text{rank} \begin{bmatrix} U_0 \\ \hat{X}_0 \end{bmatrix} = n + m. \quad (7)$$

In the absence of quantization, it is known from [5] that this rank condition holds when the input sequence is sufficiently exciting.

Definition 1 ([5]). The input sequence $\{u(k)\}_{k=0}^{T-1}$ is *persistently exciting (PE)* of order L if the Hankel matrix

$$\mathcal{U}_{0,L,T-L+1} := \begin{bmatrix} u(0) & u(1) & \cdots & u(T-L) \\ u(1) & u(2) & \cdots & u(T-L+1) \\ \vdots & \vdots & \ddots & \vdots \\ u(L-1) & u(L) & \cdots & u(T-1) \end{bmatrix}$$

has full row rank mL .

Lemma 1 ([5, Cor. 2]). *If the LTI system (1) is controllable and the input sequence $\{u(k)\}_{k=0}^{T-1}$ is PE of order $n+1$, then*

$$\text{rank} \begin{bmatrix} U_0 \\ X_0 \end{bmatrix} = n + m. \quad (8)$$

We assume

$$T \geq m(n+1) + n, \quad (9)$$

which is necessary for $\{u(k)\}_{k=0}^{T-1}$ to be PE of order $n+1$. Theorem 2 below establishes that, under the assumptions of Lemma 1, the quantized rank condition (7) also holds when the quantization errors are sufficiently small.

B. Quantized data collection

For $k < T$, by (2), (3), and (5),

$$\|x(k+1)\|_\infty \leq \|Ax(k) + Bu(k)\| \leq \alpha\sqrt{n}E_k + \beta\bar{u}.$$

To ensure that (3) holds for all $k \leq T$, we let

$$E_{k+1} := \alpha\sqrt{n}E_k + \beta\bar{u}, \quad k = 0, 1, \dots, T-1. \quad (10)$$

To maintain a prescribed quantization error bound $\delta > 0$, we let

$$N_k := \left\lceil \frac{E_k}{\delta} \right\rceil, \quad k = 0, 1, \dots, T, \quad (11)$$

where $\lceil \cdot \rceil$ is the ceiling function. Then (4) gives

$$\|d(k)\|_\infty \leq \delta \quad \forall k \in \{0, 1, \dots, T\}. \quad (12)$$

The update rule (10) yields a bound on $\|x(k+1)\|_\infty$ that is independent of the collected data but can lead to exponential growth of E_k and N_k during this stage. A tighter update rule can be obtained by exploiting the available quantized measurement $\hat{x}(k)$ and letting

$$E_{k+1} := \alpha(\|\hat{x}(k)\| + \sqrt{n}\delta) + \beta\bar{u}, \quad k = 0, 1, \dots, T-1. \quad (13)$$

Using (4), this update also ensures that (3) holds for all $k \leq T$ and can significantly reduce the required number of quantization cells.

Although the number of quantization cells may vary during this stage, E_k and N_k can be determined at each time from their update rules and, when (13) is used, previously transmitted quantized measurements; hence they need not be communicated separately. Thus, only the quantizer output at each time needs to be transmitted, and it can be represented using a finite number of bits. The number of quantization cells remains constant during the stabilization stage in Section IV.

C. Data-driven controller synthesis

Following [28], we adopt the optimization-based framework in [8] for stabilizing unknown LTI systems in the presence of disturbances and extend it to quantized state measurements.

In the absence of quantization, a stabilizing controller can be computed by solving the following *semidefinite program* (SDP):

$$\min_{(\gamma, Q, P, L)} \gamma \quad (14a)$$

subject to

$$X_1QP^{-1}Q^\top X_1^\top - P + I \preceq 0, \quad (14b)$$

$$P \succeq I, \quad (14c)$$

$$L - U_0QP^{-1}Q^\top U_0^\top \succeq 0, \quad (14d)$$

$$X_0Q = P, \quad (14e)$$

$$\text{tr}(P) + \text{tr}(L) \leq \gamma, \quad (14f)$$

where U_0 and X_0, X_1 are the input and unquantized state data matrices introduced in Section III-A. The relevant properties are summarized in the next lemma, which follows from [8, Lem. 2 and 3].

Lemma 2. 1) *Let (γ, Q, P, L) be any feasible solution to (14), and define $K := U_0QP^{-1}$. Then $A + BK$ is Schur stable (i.e., all eigenvalues lie in the open unit disk).* 2) *Suppose that the rank condition (8) holds. For any K such that $A + BK$ is Schur stable, (14) admits a feasible solution (γ, Q, P, L) satisfying $K = U_0QP^{-1}$.*

When only quantized state measurements are available, we replace X_0, X_1 in (14) with the quantized state data matrices $\hat{X}_0, \hat{X}_1$ defined in (6) and consider the following *quantized SDP*:

$$\min_{(\gamma, Q, P, L, W)} \gamma \quad (15a)$$

subject to

$$\hat{X}_1QP^{-1}Q^\top \hat{X}_1^\top - P + I \preceq 0, \quad (15b)$$

$$P \succeq I, \quad (15c)$$

$$L - U_0QP^{-1}Q^\top U_0^\top \succeq 0, \quad (15d)$$

$$\hat{X}_0Q = P, \quad (15e)$$

$$W - QP^{-1}Q^\top \succeq 0, \quad (15f)$$

$$\text{tr}(P) + \text{tr}(L) + \omega \text{tr}(W) \leq \gamma. \quad (15g)$$

The additional variable W and parameter $\omega > 0$ introduce a regularization term motivated by robustness to disturbances; see [8, Sec. 5]. Larger values of ω encourage solutions with smaller $\text{tr}(QP^{-1}Q^\top)$, which can make the sufficient condition in Theorem 1 below easier to satisfy.

The data matrices satisfy

$$X_1 = AX_0 + BU_0, \quad \hat{X}_0 = X_0 + D_0, \quad \hat{X}_1 = X_1 + D_1.$$

Hence

$$\hat{X}_1 = AX_0 + BU_0 + D_1 = A\hat{X}_0 + BU_0 + (D_1 - AD_0).$$

Let

$$\Delta_0 := D_1 - AD_0 \in \mathbb{R}^{n \times T},$$

which captures the discrepancy introduced by the quantized measurements. By (2) and (12), the columns of Δ_0 satisfy

$$\|d(k+1) - Ad(k)\| \leq \sqrt{n}(\alpha+1)\delta \quad \forall k \in \{0, 1, \dots, T-1\}.$$

Hence

$$\|\Delta_0\| \leq \|\Delta_0\|_F \leq \sqrt{nT}(\alpha + 1)\delta =: \hat{\delta}, \quad (16)$$

where $\hat{\delta}$ can be made arbitrarily small by decreasing δ , at the expense of requiring more quantization cells.

The following theorem from [28] establishes a verifiable condition under which any feasible solution to the SDP (15) yields a stabilizing controller.

Theorem 1 ([28, Th. 1]). *Let $(\bar{\gamma}, \bar{Q}, \bar{P}, \bar{L}, \bar{W})$ be any feasible solution to (15), and define*

$$\bar{K} := U_0 \bar{Q} \bar{P}^{-1}, \quad \bar{M} := \bar{Q} \bar{P}^{-1} \bar{Q}^\top.$$

Then

$$(A + B\bar{K})\bar{P}(A + B\bar{K})^\top - \bar{P} \preceq -\eta_1 I, \quad (17)$$

where

$$\eta_1 := 1 - \hat{\delta}^2 \|\bar{M}\| - 2\hat{\delta} \|\hat{X}_1 \bar{M}\|. \quad (18)$$

If

$$\eta_1 > 0, \quad (19)$$

then $A + B\bar{K}$ is Schur stable.

The following theorem guarantees that the verifiable condition (19) is satisfied by every optimal solution when the quantization errors are sufficiently small.

Theorem 2. *Suppose that the LTI system (1) is controllable and the input sequence $\{u(k)\}_{k=0}^{T-1}$ is PE of order $n + 1$. There exists $\delta^* > 0$ such that, if the quantization error bound $\delta < \delta^*$, then the rank condition (7) holds, the quantized SDP (15) is feasible, and every optimal solution to (15) satisfies (19).*

A related result was established in [28, Th. 2], which ensures that every optimal solution yields a stabilizing controller for sufficiently small quantization errors. Theorem 2 further guarantees that every optimal solution satisfies the verifiable condition (19), but under a more restrictive sufficient bound on δ .

Theorem 2 is restricted to optimal solutions to the quantized SDP (15), in contrast to Theorem 1, which applies to all feasible solutions. This restriction arises because the proof relies on comparing the optimal value of (15) with the objective value of a specifically constructed feasible solution.

Due to space limitation, the proof of Theorem 2 is omitted here and can be found in [29]. It uses the robust fundamental lemma in [30] and includes an explicit sufficient bound on δ . The bound depends on the unknown matrices A and B and therefore cannot be verified directly from the available data. Nevertheless, it provides an existence guarantee: the verifiable condition (19) can always be satisfied by sufficiently fine quantization.

IV. QUANTIZED FEEDBACK STABILIZATION

Having obtained a stabilizing feedback gain from quantized data, we now address closed-loop stabilization using quantized state feedback.

A. Quantized state feedback implementation

We first recall the quantized state feedback implementation introduced in [28]. At time T , suppose that a feasible solution $(\bar{\gamma}, \bar{Q}, \bar{P}, \bar{L}, \bar{W})$ to the SDP (15) has been obtained and satisfies (19). Then by Theorem 1, the feedback gain $\bar{K} := U_0 \bar{Q} \bar{P}^{-1}$ satisfies (17), and $A + B\bar{K}$ is Schur stable.

For the transition between the two stages, we let $u(T) = 0$ and begin quantized state feedback at time $T + 1$. During the stabilization stage, the quantizer uses a fixed number of quantization cells and exponentially decaying bounds:

$$N_k = N_s, \quad E_{k+1} = \mu E_k, \quad k = T + 1, T + 2, \dots, \quad (20)$$

where the initial bound E_{T+1} , the contraction factor μ , and a sufficient lower bound on N_s for $\mu \in (0, 1)$ are derived in Section IV-B. With these choices, the closed-loop state under

$$x(k + 1) = Ax(k) + B\bar{K}\hat{x}(k), \quad k \geq T + 1, \quad (21)$$

remains in the quantization range (i.e., (3) holds) for all $k \geq T + 1$ and converges exponentially to the origin.

During this stage, since the number of quantization cells remains constant, the quantizer can be expressed as a “zooming” quantizer [12]:

$$\hat{x}(k) = \mu_k q\left(\frac{x(k)}{\mu_k}\right),$$

where $q(\cdot)$ is the quantizer with bound E_{T+1} and N_s quantization cells per dimension, and

$$\mu_k := \mu^{k-T-1}, \quad k = T + 1, T + 2, \dots$$

is the zoom variable.

B. Exponential convergence

We establish exponential convergence by constructing nested sets that contract under the closed-loop dynamics, based on sublevel sets of

$$V(x) := x^\top \bar{P}^{-1} x, \quad (22)$$

where $\bar{P} \succ 0$ is from the feasible solution to (15) obtained at time T .

We first construct a sublevel set of V containing $x(T + 1)$ and derive E_{T+1} such that the quantization range $\mathcal{R}(E_{T+1})$ contains this set. Recall that

$$\mathcal{R}(E) = \{x \in \mathbb{R}^n : \|x\|_\infty \leq E\},$$

and define the sublevel set

$$\mathcal{S}(E) := \left\{x \in \mathbb{R}^n : V(x) \leq \frac{E^2}{\bar{\lambda}(\bar{P})}\right\}.$$

Since

$$\frac{\|x\|_\infty^2}{\bar{\lambda}(\bar{P})} \leq \frac{\|x\|^2}{\bar{\lambda}(\bar{P})} \leq V(x) \leq \frac{\|x\|^2}{\underline{\lambda}(\bar{P})} \leq \frac{n\|x\|_\infty^2}{\underline{\lambda}(\bar{P})} \quad \forall x \in \mathbb{R}^n,$$

we have

$$\mathcal{S}(E) \subset \mathcal{R}(E) \subset \mathcal{S}\left(\sqrt{\frac{n\bar{\lambda}(\bar{P})}{\underline{\lambda}(\bar{P})}} E\right) \quad \forall E > 0.$$

Using $u(T) = 0$ and $x(T) \in \mathcal{R}(E_T)$, we obtain

$$\|x(T+1)\|_\infty \leq \|Ax(T)\| \leq \alpha\|x(T)\| \leq \alpha\sqrt{n}E_T.$$

Therefore, defining

$$E_{T+1} := \alpha n \sqrt{\frac{\bar{\lambda}(\bar{P})}{\underline{\lambda}(\bar{P})}} E_T \quad (23)$$

yields $x(T+1) \in \mathcal{R}(\alpha\sqrt{n}E_T) \subset \mathcal{S}(E_{T+1}) \subset \mathcal{R}(E_{T+1})$. Thus, $\mathcal{R}(E_{T+1})$ can be used as the initial quantization range for the stabilization stage.

A tighter initial bound can be obtained by exploiting the available quantized measurement $\hat{x}(T)$ and letting

$$E_{T+1} := \alpha \sqrt{\frac{n \bar{\lambda}(\bar{P})}{\underline{\lambda}(\bar{P})}} (\|\hat{x}(T)\| + \sqrt{n}\delta), \quad (24)$$

which also ensures $x(T+1) \in \mathcal{S}(E_{T+1}) \subset \mathcal{R}(E_{T+1})$.

The following theorem makes the sufficient condition for exponential convergence in [28, Th. 3] more explicit by introducing an ε -independent bound N^* on the number of quantization cells. This formulation facilitates the analysis of the quantization trade-off in Section V. For convenience, define

$$\theta := \frac{(\alpha + \beta\|\bar{K}\|)\beta\|\bar{K}\|\bar{\lambda}(\bar{P})}{\underline{\lambda}(\bar{P})}, \quad \phi := \frac{\beta^2\|\bar{K}\|^2}{\underline{\lambda}(\bar{P})}, \quad (25)$$

where $\alpha, \beta > 0$ are as in (2).

Theorem 3. *Suppose that the number of quantization cells satisfies*

$$N_s > \frac{\theta + \sqrt{\theta^2 + \eta_1\phi}}{\eta_1} \sqrt{\frac{n \bar{\lambda}(\bar{P})^3}{\underline{\lambda}(\bar{P})}} =: N^*, \quad (26)$$

where $\eta_1 \in (0, 1)$ is defined in (18). Then there exists $\varepsilon > 0$ such that

$$N_\varepsilon^* := \sqrt{\frac{(\theta + \sqrt{\theta^2 + \eta_1(\phi + \varepsilon)})^2 n \bar{\lambda}(\bar{P})^3}{\eta_1^2 \underline{\lambda}(\bar{P})} - \varepsilon n \bar{\lambda}(\bar{P})} \quad (27)$$

satisfies $N_s > N_\varepsilon^* > N^*$. Moreover, for any such ε ,

$$\|x(k)\|_\infty \leq \mu^{k-T-1} E_{T+1} \quad \forall k \geq T+1, \quad (28)$$

where

$$\mu := \sqrt{\max\left\{\frac{(N_\varepsilon^*)^2}{N_s^2}, 1 - \frac{n\varepsilon \bar{\lambda}(\bar{P})}{N_s^2}\right\}} < 1. \quad (29)$$

The proof of Theorem 3 relies on the following lemma.

Lemma 3. *For any $k \geq T+1$, if*

$$x(k) \in \mathcal{S}(E_k),$$

then

$$x(k+1) \in \mathcal{S}(\mu E_k),$$

where μ is defined in (29).

Due to space limitation, the proof of Lemma 3 is omitted here and can be found in [29].

Proof of Theorem 3. First, note that N_ε^* is well defined and positive for every $\varepsilon > 0$, since $(\theta + \sqrt{\theta^2 + \eta_1(\phi + \varepsilon)})^2 \geq \eta_1\varepsilon$ and $\bar{\lambda}(\bar{P})^2 > \eta_1 \underline{\lambda}(\bar{P})$. Moreover, N_ε^* is strictly increasing in ε , and

$$\lim_{\varepsilon \searrow 0} N_\varepsilon^* = N^*.$$

Thus, under (26), there exists a sufficiently small $\varepsilon > 0$ such that $N_s > N_\varepsilon^* > N^*$. For any such ε , both terms in the maximum in (29) are strictly smaller than one, and $N_\varepsilon^* > 0$. Hence $\mu \in (0, 1)$.

By either (23) or (24),

$$x(T+1) \in \mathcal{S}(E_{T+1}).$$

Therefore, iterating Lemma 3 gives

$$x(k) \in \mathcal{S}(\mu^{k-T-1} E_{T+1}) \subset \mathcal{R}(\mu^{k-T-1} E_{T+1})$$

for all $k \geq T+1$, which proves (28). $\square$

V. TRADE-OFF BETWEEN DESIGN AND STABILIZATION QUANTIZATION REQUIREMENTS

The proposed approach involves different quantization requirements for controller design and subsequent stabilization. In the design stage, a smaller quantization error bound generally requires more quantization cells, but can improve the verifiable condition (19) and reduce the sufficient bound N^* in (26) for exponential convergence in the stabilization stage. In this section, we characterize this trade-off through the relations among the design-stage quantization error bound δ , the number of quantization cells per dimension N_k during data collection, the resulting coefficient η_1 in Theorem 1, and the sufficient bound N^* in Theorem 3.

We use the following standard one-sided asymptotic notation. For two functions $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$, we write $f(s) = O(g(s))$ as $s \rightarrow 0^+$ if there exist $C, s_0 > 0$ such that

$$|f(s)| \leq Cg(s) \quad \forall s \in (0, s_0).$$

If f is also nonnegative, then we write $f(s) = \Omega(g(s))$ if $g(s) = O(f(s))$, and $f(s) = \Theta(g(s))$ if both $f(s) = O(g(s))$ and $f(s) = \Omega(g(s))$.

We first characterize the quantization requirements in both stages for sufficiently fine quantization in the design stage.

Proposition 1. *Suppose that the LTI system (1) is controllable, and consider a fixed initial state x_0 and a fixed input sequence $\{u(k)\}_{k=0}^{T-1}$ that is PE of order $n+1$. As $\delta \rightarrow 0^+$,*

$$N_k = \Theta(\delta^{-1}) \quad \forall k \in \{0, 1, \dots, T\}.$$

Moreover, uniformly over all optimal solutions $(\bar{\gamma}, \bar{Q}, \bar{P}, \bar{L}, \bar{W})$ to the quantized SDP (15),

$$1 - \eta_1 = O(\delta), \quad N^* = O(1).$$

Proof. For the fixed initial state and input sequence, the unquantized state sequence during the design stage is independent of δ . By (12), the quantized state sequence and corresponding data matrix satisfy

$$\|\hat{x}(k) - x(k)\| \leq \sqrt{n}\|d(k)\|_\infty = O(\delta)$$

for all $k \in \{0, 1, \dots, T\}$, and

$$\|\hat{X}_1 - X_1\| \leq \|D_1\|_F = O(\delta).$$

We first characterize N_k for $k \leq T$. Under the update rule (10), every $E_k > 0$ is independent of δ . For the measurement-dependent update rule (13), E_0 is fixed, and

$$\begin{aligned} E_k &= \alpha(\|\hat{x}(k-1)\| + \sqrt{n}\delta) + \beta\bar{u} \\ &= \alpha(\|x(k-1)\| + O(\delta)) + \beta\bar{u} \\ &= \alpha\|x(k-1)\| + \beta\bar{u} + O(\delta) \end{aligned}$$

for all $k \in \{1, 2, \dots, T\}$. Since the unquantized state sequence is independent of δ and $\alpha\|x(k-1)\| + \beta\bar{u} > 0$, we have $E_k = \Theta(1)$ for every $k \leq T$. Therefore, under either (10) or (13), (11) implies

$$N_k = \left\lceil \frac{E_k}{\delta} \right\rceil = \frac{E_k}{\delta} + O(1) = \Theta(\delta^{-1})$$

for all $k \in \{0, 1, \dots, T\}$.

Changes in δ may lead to different quantized measurements and consequently different solutions to the quantized SDP (15). Nevertheless, the proof of Theorem 2 in [29] shows that, as $\delta \rightarrow 0^+$, the following relations hold uniformly over all optimal solutions to (15):

$$\|\bar{K}\| = O(1), \quad \|\bar{M}\| = O(1), \quad 1 \leq \underline{\lambda}(\bar{P}) \leq \bar{\lambda}(\bar{P}) = O(1).$$

We now characterize $1 - \eta_1$. Since $\hat{\delta} = \sqrt{nT}(\alpha + 1)\delta$, $\|\bar{M}\| = O(1)$, and

$$\|\hat{X}_1 \bar{M}\| \leq (\|X_1\| + O(\delta))O(1) = O(1),$$

(18) implies

$$1 - \eta_1 = O(\delta^2)O(1) + O(\delta)O(1) = O(\delta).$$

Finally, (25) yields

$$\theta = O(1), \quad \phi = O(1).$$

Moreover, $1 - \eta_1 = O(\delta)$ implies that, for sufficiently small δ , there exists a positive lower bound on η_1 , uniformly over all optimal solutions. Therefore, (26) implies

$$N^* \leq \frac{2\theta + \sqrt{\phi}}{\eta_1} \sqrt{\frac{n \bar{\lambda}(\bar{P})^3}{\underline{\lambda}(\bar{P})}} = O(1). \quad \square$$

We next characterize the growth of the sufficient bound N^* as $\eta_1 \rightarrow 0^+$.

Proposition 2. Suppose that $\rho(A) > 1$. Along any sequence of feasible solutions $(\bar{\gamma}, \bar{Q}, \bar{P}, \bar{L}, \bar{W})$ to the quantized SDP (15) such that $\eta_1 > 0$ and $\eta_1 \rightarrow 0^+$,

$$N^* = \Omega(\eta_1^{-1}). \quad (30)$$

Proof. If $\eta_1 > 0$, then Theorem 1 implies that $A + B\bar{K}$ is Schur stable. Define

$$d_A := \inf_{\rho(F) < 1} \|A - F\|.$$

Since $\rho(A) > 1$ and the spectral radius is continuous, $d_A > 0$. Moreover

$$d_A \leq \|A - (A + B\bar{K})\| = \|B\bar{K}\| \leq \beta\|\bar{K}\|.$$

Since $\bar{\lambda}(\bar{P}) \geq \underline{\lambda}(\bar{P}) \geq 1$, (25) yields

$$\theta \geq (\alpha + \beta\|\bar{K}\|)\beta\|\bar{K}\| \geq (\alpha + d_A)d_A.$$

Therefore, (26) implies

$$N^* \geq \frac{2\theta\sqrt{n}}{\eta_1} \geq \frac{2(\alpha + d_A)d_A\sqrt{n}}{\eta_1} = \Omega(\eta_1^{-1}). \quad \square$$

Remark 1. If $\|\bar{K}\| = O(1)$ and $\bar{\lambda}(\bar{P}) = O(1)$ along the sequence of feasible solutions in Proposition 2, then (25) yields $\theta = O(1)$ and $\phi = O(1)$. Consequently, since $\underline{\lambda}(\bar{P}) \geq 1$, (26) yields $N^* = O(\eta_1^{-1})$. Together with (30), this gives

$$N^* = \Theta(\eta_1^{-1}).$$

The trade-off characterized in Propositions 1 and 2 is illustrated numerically in Section VI.

VI. NUMERICAL ILLUSTRATION

A. Quantized controller design and stabilization

We illustrate the proposed approach on the LTI system (1) with

$$A = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

which is unstable in open loop. The dimensions are $n = 2$ and $m = 1$. For the controller design stage, we use $T = m(n+1) + n = 5$, the minimum data length allowed by (9). The stabilization stage is simulated for 30 steps.

During the design stage, the input sequence is PE of order $n+1 = 3$ and satisfies $\|u(k)\| \leq \bar{u} = 1$. The initial state is $x_0 = [1 \ 1]^\top$. We use $E_0 = 1$, $\delta = 0.01$, $\alpha = 3 > \|A\|$, and $\beta = 1.5 > \|B\|$. Using (11) together with the measurement-dependent update rule (13), the number of quantization cells per dimension is

$$(N_0, N_1, \dots, N_5) = (100, 461, 1174, 951, 1742, 2658).$$

The collected data satisfy the rank condition (7). Solving the quantized SDP (15) with $\omega = 1$ yields

$$\bar{K} = U_0 \bar{Q} \bar{P}^{-1} = \begin{bmatrix} -1.129 & -0.333 \end{bmatrix},$$

and $\eta_1 = 0.338 > 0$. Thus, the verifiable condition (19) holds, and Theorem 1 guarantees that $A + B\bar{K}$ is Schur stable. Indeed, its eigenvalues are 0.392 and -0.521 .

During the stabilization stage, the initial quantization range is computed using the measurement-dependent initialization (24), yielding $E_{T+1} = 118.9$. The sufficient condition (26) in Theorem 3 gives the bound $N^* = 2623.6$ and hence $N_s \geq 2624$. In the simulation, we use only $N_s = 50$ cells per dimension and set $\mu = 0.95$. Although these choices are not certified by Theorem 3, the simulation results in Fig. 1 show successful stabilization.

Figure 1(a) shows that the state remains within the quantization range and converges to the origin. During the design stage ($k = 0, \dots, 5$), the quantizer maintains the prescribed quantization error bound. At the transition to stabilization, the quantization range increases to $E_{T+1} = 118.9$ according to (24), and thereafter decreases exponentially with $\mu = 0.95$.

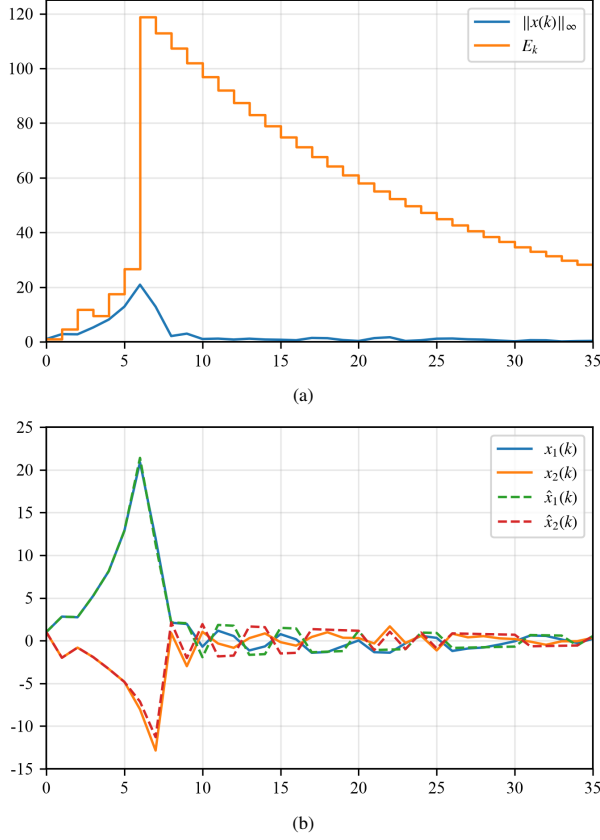


Fig. 1: Evolution of (a) the state norm $\|x(k)\|_\infty$ and quantization bound E_k , and (b) the state components $x_i(k)$ and quantized measurements $\hat{x}_i(k)$. The design stage ends at $k = 5$, and quantized state feedback begins at $k = 6$.

Figure 1(b) shows that the quantized measurements closely track the states during data collection, and both converge during stabilization.

The sufficient bound in Theorem 3 is conservative in this example. Although it requires $N_s \geq 2624$, convergence is achieved with $N_s = 50$ in simulation. This conservativeness is due partly to the worst-case norm bounds in the analysis and the conversion between the hypercubic quantization ranges and the Lyapunov sublevel sets. More careful analysis may substantially reduce the quantization requirement.

B. Quantization trade-off

We next illustrate the trade-off characterized in Section V. We vary the design-stage quantization error bound δ while keeping the initial condition, input sequence, and other parameters fixed. For each value of δ , we generate the quantized state measurements, formulate and solve the corresponding quantized SDP (15), and compute the resulting coefficient η_1 in Theorem 1 and sufficient bound N^* in Theorem 3. The results are shown in Fig. 2 and Table I.

Figure 2(a) shows that η_1 approaches 1 as δ decreases. This is consistent with $1 - \eta_1 = O(\delta)$ as $\delta \rightarrow 0^+$ established in Proposition 1. On the other hand, η_1 decreases as δ increases in this example and becomes negative between $\delta = 0.015$ and 0.0154. For the larger values of δ considered beyond this crossing, the condition $\eta_1 > 0$ is no longer satisfied.

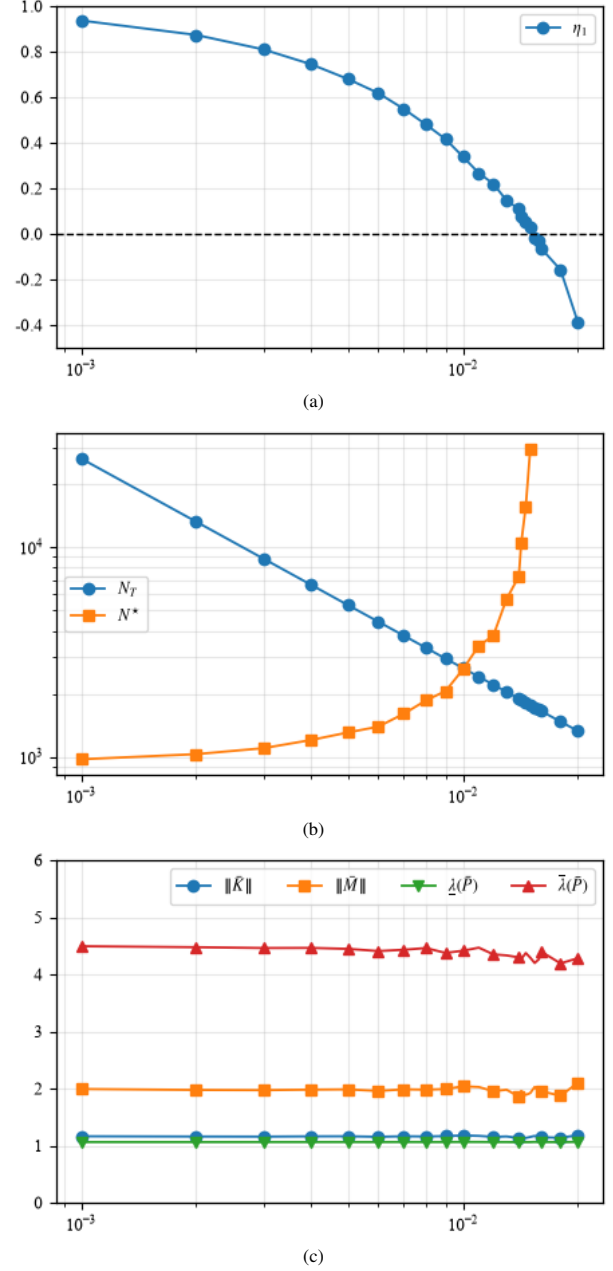


Fig. 2: Trade-off between design and stabilization quantization requirements as δ varies: (a) the coefficient η_1 in the verifiable condition (19), (b) the number of quantization cells per dimension N_T and the sufficient bound N^* for exponential convergence, and (c) $\|K\|$ and $\|M\|$ and the smallest and largest eigenvalues of P for the optimal solutions to (15). The bound N^* is shown only when $\eta_1 > 0$.

TABLE I: Quantization trade-off for different values of δ

δ	N_T	η_1	N^*
0.001	26521	0.9367	976.8
0.005	5304	0.6796	1311.4
0.010	2658	0.3385	2623.6
0.014	1900	0.1091	7270.7
0.015	1773	0.0273	29572.5
0.0154	1725	-0.0172	—

Figure 2(b) illustrates the corresponding trade-off between the two stages. Over the range considered, N_T increases approximately inversely with δ , while N^* decreases as δ decreases. For example, decreasing δ from 0.01 to 0.001 increases N_T from 2658 to 26521, while N^* decreases from approximately 2624 to 977. In the opposite direction, increasing δ reduces the design-stage quantization requirement but causes η_1 to approach zero and N^* to increase rapidly. For example, at $\delta = 0.015$, N_T has decreased to 1773, while $\eta_1 = 0.0273$ and $N^* \approx 2.96 \times 10^4$. These behaviors are consistent with Propositions 1 and 2.

Figure 2(c) shows that $\|\bar{K}\|$, $\|\bar{M}\|$, $\underline{\lambda}(\bar{P})$, and $\bar{\lambda}(\bar{P})$ vary relatively mildly over the range considered, in contrast to the rapid growth of N^* as η_1 approaches zero. In particular, the observed behavior of $\|\bar{K}\|$ and $\bar{\lambda}(\bar{P})$ is consistent with the additional boundedness conditions in Remark 1, under which the asymptotic relation can be strengthened to $N^* = \Theta(\eta_1^{-1})$.

Overall, the example illustrates that finer quantization in the design stage requires more quantization cells for data collection, but can substantially reduce the sufficient bound for exponential convergence in the stabilization stage.

VII. CONCLUSION

This paper studied data-driven stabilization of unknown discrete-time linear systems using only quantized state measurements that take values in finite sets. Building on a two-stage framework for controller design and feedback stabilization, we strengthened the controller-design guarantee, derived a more explicit sufficient condition for exponential convergence, and characterized the trade-off between the quantization requirements in the two stages. Numerical results illustrate this trade-off and the conservativeness of the theoretical sufficient bound.

Future work includes reducing the conservativeness of the quantizer design and extending the proposed framework to continuous-time LTI systems and switched systems.

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